

Team Worksheet — Lab 15

Variation, Run Charts, Pareto and Stratification — In Depth

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in lab-15-run-chart-pareto-stratification.xlsx.

3. What we produced

Deliverable: A run chart with pattern analysis, a Pareto chart and a stratified defect analysis.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Recap common versus special cause variation in full, with the management response each demands — process change versus local investigation.

Step 2. Understand tampering (Deming's funnel): adjusting a stable process in response to common-cause noise increases variation. Demonstrate it with the bead or funnel experiment.

Step 3. Plot your capstone Y as a run chart over time, with the median as the centre line.

Step 4. Recap and apply the six run chart patterns in full: trend, shift, cluster, mixture, oscillation and bias — with the run rules that detect each.

Step 5. Mark any special-cause signals and investigate what changed at that point in time. Special causes must be explained, not averaged away.

Step 6. Recap the Pareto principle and build a Pareto chart of defect categories for your capstone, with the cumulative percentage line.

Step 7. Identify the vital few categories accounting for roughly 80 percent of the impact. Weight by cost, not count, where defect costs differ materially.

Step 8. Now stratify. Re-draw the Pareto separately by plant, line, shift, operator, product family and supplier.

Step 9. Look for a stratification factor where the Pareto shape changes dramatically — that factor is a strong candidate X and tells you where to focus.

Step 10. Build boxplots of your Y across each stratification factor to compare medians, spread and outliers visually before testing statistically.

Step 11. Record your top three candidate Xs from stratification — these feed directly into the hypothesis tests in Lab 17.

7. Self-check

Your run chart is assessed against all six patterns, your Pareto is weighted appropriately, and stratification has produced at least three named candidate Xs for statistical testing.

Complete? ☐ Yes ☐ Not yet

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